$$2KClO_3 + 6MnO_2 \rightarrow 6MnO_3 + 2KCl$$

It is an intermediate.

$$6MnO_3 \rightarrow 6MnO_2 + 3O_2$$

**Example 3:**

Formation of water due to the reaction of $H_2$ and $O_2$ in the presence of Cu proceeds as follows. Steps in the reaction $H_2 + ½O_2 \rightarrow H_2O$ can be given as

$$2Cu + \frac{1}{2}O_2 \rightarrow Cu_2O$$

It is an intermediate.

$$Cu_2O + H_2 \rightarrow H_2O + 2Cu$$

**Example 4**:

Oxidation of HCl by air in presence of $CuCl_2$ proceeds as follows. Steps in the reaction $4HCl + O_2 \rightarrow 2H_2O + 2Cl_2$ can be given as

$$2CuCl_2 \rightarrow Cl_2 + Cu_2Cl_2$$

$$2Cu_2Cl_2 + O_2 \rightarrow 2Cu_2OCl_2$$

It is an intermediate.

$$2Cu_2OCl_2 + 4HCl \rightarrow 2H_2O + 4CuCl_2$$

This theory describes

(i) the specificity of a catalyst and

(ii) the increase in the rate of the reaction with increase in the concentration of a catalyst.

### Limitations

(i) The intermediate compound theory fails to explain the action of catalytic poison and activators (promoters).

(ii) This theory is unable to explain the mechanism of heterogeneous catalysed reactions.

### 2. Adsorption theory

Langmuir explained the action of catalyst in heterogeneous catalysed reactions based on adsorption. The reactant molecules are adsorbed on the catalyst surfaces, so this can also be called as contact catalysis.

According to this theory, the reactants are adsorbed on the catalyst surface to form an activated complex which subsequently decomposes and gives the product.

The various steps involved in a heterogeneous catalysed reaction are given as follows:

1. Reactant molecules diffuse from bulk to the catalyst surface.
2. The reactant molecules are adsorbed on the surface of the catalyst.
3. The adsorbed reactant molecules are activated and form activated complex which is decomposed to form the products.
4. The product molecules are desorbed.
5. The product diffuse away from the surface of the catalyst.



XII U10-Surface Chemistry.indd 81 11-01-2021 15:33:34